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A physics-based compact model for MoS_2 field-effect transistors considering the band-tail effect and contact resistance

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In this paper, we present a compact surface-potential-based drain current model in molybdenum disulfide (MoS_2) field-effect transistors (FETs). Considering variable range hopping (VRH) transport via band-tail states in MoS_2 transistors, an explicit solution for surface potential has been derived and it provides a good description over different regions of operation by comparisons with numerical data. Based on the charge-sheet model (CSM), which applies to drift-diffusion transport, the current expression including contact resistance and velocity saturation effect is developed. Furthermore, the presented model is validated and shows a good agreement with the experimental data for MoS_2 FETs.
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1. Introduction

For the post-silicon era, 2D layered transition metal dichalcogenide (TMD) has become attractive for use in next-generation electronics because of its excellent scalability.[1,2] Among them, the single-atomic-layer molybdenum disulfide (MoS_2) has shown promising prospects as a channel material for field-effect transistors (FETs). Some complex digital logic and high-frequency ac applications on MoS_2 FETs such as static random access memories and five-stage ring oscillators have been observed experimentally.[3] Thus, the demand for an accurate and better understanding compact model for the 2D FETs is really an urgent task at the new circuit-level applications.

Previously developed compact models[4–7] have been proposed for current–voltage (I–V) characteristics of TMD FETs based on the drift-diffusion picture. In the beginning, a model[4] based on the capacitance network was developed. It only described the intrinsic device characteristics and excluded the band-tail effect and the diffusion component in the current calculation. Considering Fermi–Dirac statistics, a workable model[5] for TMD-based FETs was proposed, but it did not account for the trap charge and involved implicit calculation. To avoid an iterative solution, the model in Ref. 6 got an explicit I–V expression, but when solving the surface potential, it only invoked the free charge and ignored the influence of the localized states. The trap density at room temperature was estimated at the actual manufacturing process of MoS_2 transistors.[8] Electrical conduction occurs via localized states, which is often described by variable range hopping (VRH) transport and should be taken into account for a complete model. To better reflect a practical MoS_2 FET, Ref. 7 included trap charge in the calculation of surface potential through Taylor expansion, but this calculation process was complicated and the accuracies were limited (maximum absolute error in the mV range). Recently, for short channel devices, a recent finding with the aid of atomistic simulations has been reported,[9] which accurately reproduced the gate control of the MoS_2 FETs in both long and short channel transistors.

To this end, in this work, through a regional approach, an explicit asymptotic solution of surface potential for MoS_2 FETs is derived based on the exponential localized-state distribution. Comparison with the numerical data shows that the closed-form solution serves as a good approximation to potential under different voltage conditions. The drain current model based on surface potential is obtained. Two issues of great importance in the emerging 2D FET arena, that is, contact resistance and velocity saturation, have been carefully considered. Furthermore, the calculated drain current is compared with experimental results to ensure the model's accuracy.

2. Surface potential calculation

As shown in Fig. 1, due to the ultra-thin channels for MoS_2 devices, a uniform electrostatic potential of the 2D layer is assumed.[9] The effective potential energy is an equipotential which is also called "channel potential",[9] and no bending normal to the channel direction. Note that, in order to facilitate the connection with previous publications,[6,7] a familiar concept of "surface potential" is also used in this paper to represent the effective channel potential that behaves as a whole from the electrostatic perspective. Furthermore, for MoS_2 transistors, the band-tail effect has a significant impact on the characteristics.[10] An exponential distribution of trap density is considered for the modeling, and the modified total density of states (DOS) expression is represented as a function of the energy E:[7]

$$\mathrm{DOS}_{2\mathrm{D}}(E) = \begin{cases} D_0 \ \text{ above } E_C \\ N_T \exp\left(\dfrac{E - E_C}{k_\mathrm{B} T_0}\right) \text{band tail below } E_C \end{cases} \tag{1}$$

where D_0 [$\mathrm{cm}^{-2}\,\mathrm{eV}^{-1}$] is the constant DOS; N_T [$\mathrm{cm}^{-2}\,\mathrm{eV}^{-1}$] denotes the total concentration of trap states; E_C is the conduction band energy level; k_B and T_0 are the Boltzmann constant and the characteristic temperature of the exponential DOS, respectively. Using DOS and Fermi statistics $f(E - E_f) = \left[1 + \exp\left(\frac{E - E_f}{k_\mathrm{B} T}\right)\right]^{-1}$ with T as operation temperature, the total carrier number density is given by the sum of the trap (n_trap) and free (n_free) carrier number

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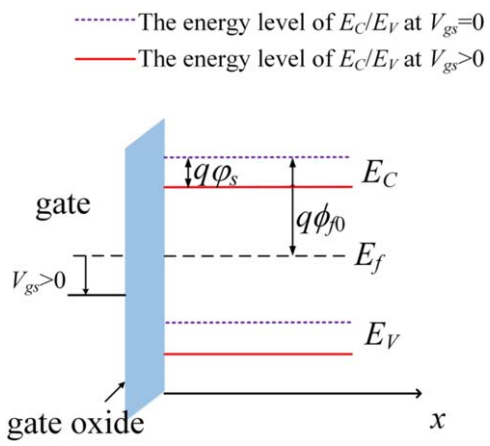

Fig. 1. (Color online) Energy band diagram along the vertical channel direction (x-axis). E_f and φ_s represent Fermi energy and surface potential, respectively.

densities can be obtained after integrating Eq. (1):

$$n_{\text{total}} = n_{\text{trap}} + n_{\text{free}} = N_T\theta_t \exp((\varphi_s + \phi_{f0} - V)/E_1) + D_0 k_B T \exp((\varphi_s + \phi_{f0} - V)/\phi_{\text{th}}) \quad (2)$$

where $\theta_t = \pi k_B T/\sin(\pi T/T_0)$; $E_1 = k_B T_0/q$; ϕ_{th} is the thermal voltage; V is the quasi-Fermi potential; and ϕ_{f0} is the potential gap between E_C and the Fermi level at a neutral condition. Furthermore, based on Gauss's law,[11] the surface potential can be calculated by the implicit relation:

$$Q_s = C_{\text{OX}}(V_{\text{gb}} - \varphi_s) = qN_T\theta_t \exp((\varphi_s + \phi_{f0} - V)/E_1) + qD_0 k_B T \exp((\varphi_s + \phi_{f0} - V)/\phi_{\text{th}}) \quad (3)$$

where Q_s is charge density; $V_{\text{gb}} = V_{\text{gs}} - V_{\text{fb}}$; V_{fb} is the flat-band voltage, and C_{OX} is the gate oxide capacitance. However, the calculation of φ_s from (3) requires time-consuming numerical simulations. To solve this problem, the regional approach is adopted here to derive the explicit approximation of φ_s.

In the subthreshold region, trap states play the central role and the free charge term can be neglected. Consequently, the surface potential (φ_{sub}) only accounting for the trap states can be derived via the Lambert W function (W_0):[12]

$$\varphi_{\text{sub}} = E_1 \ln\left[W_0\left(\frac{A_0}{E_1}\exp\left(\frac{V_{\text{gb}}}{E_1}\right)\right)\left(\frac{E_1}{A_0}\right)\right] \quad (4)$$

where $A_0 = (qN_T\theta_t/C_{\text{OX}}) \cdot \exp((\phi_{f0} - V)/E_1)$.

In the case of the above-threshold regime, free charge dominates Gauss's law. Accordingly, one can find another asymptotic equation for the surface potential in the above-threshold region (φ_{ab}), i.e.,

$$\varphi_{\text{ab}} = \phi_{\text{th}} \ln\left[W_0\left(\frac{A_1}{\phi_{\text{th}}}\exp\left(\frac{V_{\text{gb}}}{\phi_{\text{th}}}\right)\right)\left(\frac{\phi_{\text{th}}}{A_1}\right)\right] \quad (5)$$

where $A_1 = (qk_B T D_0/C_{\text{OX}}) \cdot \exp((\phi_{f0} - V)/\phi_{\text{th}})$.

A continuous transition from the subthreshold to above-threshold regions can be acquired by a smooth function as in Ref. 13 and a unified surface potential solution (φ_{s0}) can be obtained. Similar to Ref. 6, the values of surface potential at the source end (φ_{ss}) and drain end (φ_{sd}) are solved by using $V = 0$ and $V = V_{\text{ds}}$ in (3), respectively. Furthermore, it should be noted that the accuracy of the obtained surface potential can be improved by adding a correction term[7] as follows:

$$\varphi_s = \varphi_{s0} + \omega(y_\omega, y'_\omega, y''_\omega) \quad (6)$$

where

$$\omega(y_\omega, y'_\omega, y''_\omega) = -\frac{y_\omega}{y'_\omega}\left(1 + \frac{y''_\omega y_\omega}{2y'_\omega y'_\omega}\right). \quad (7)$$

In (7), y_w' and y_w'' are the first and second derivatives of y_w, respectively, where y_w is

$$y_\omega = C_{\text{OX}}(V_{\text{gb}} - \varphi_s) - q(n_{\text{trap}} + n_{\text{free}}). \quad (8)$$

3. Charge-sheet model based on the proposed surface potential solution

Once the surface potential is solved precisely, a drain current model can be expressed based on the charge-sheet model (CSM) as Ref. 14

$$I_{\text{ds}} = \mu_{\text{eff}} W Q_s \frac{dV}{dy} = \mu_{\text{eff}} W Q_s \frac{dV}{d\varphi_s - V}\frac{d\varphi_s - V}{dy} \quad (9)$$

where W is the device width; y is the coordinate along the channel; and μ_{eff} is the effective mobility.

The conduction mechanism of VRH plays a significant role in the mobility of MoS_2, which also depends on the surface potential. Therefore, similar to Ref. 7, we determine the effective mobility μ_{eff} under VRH as

$$\mu_{\text{eff}} = \mu_0 + \mu_{\text{hop}} \exp\left(\frac{\varphi_s - \varphi_{\text{st}} - (E_{\text{tr}}/q)}{\phi_{\text{th}}}\right). \quad (10)$$

Here, μ_0 is the channel mobility in the subthreshold region; μ_{hop} is the hopping mobility, and E_{tr} is the effective transport energy which gets most frequently visited by charge carrier via localized states.[7]

We define φ_{st} as the surface potential in (3) corresponding to $V_{\text{gs}} = V_{\text{T}}$. Herein, V_{T} represents the threshold voltage which can be calculated by the condition that $n_{\text{free}} = n_{\text{trap}}$. As a result, we can obtain:

$$\varphi_{\text{st}} = -\phi_{f0} + \phi_{\text{th}} \ln\left[\left(\frac{D_0 k_B T}{N_T\theta_t}\right)^{\frac{T_0}{T - T_0}}\right]. \quad (11)$$

According to (10), we can find that μ_0 dominates in the subthreshold region, and in the above-threshold regime, the exponential term in (10) becomes the dominant term.

Furthermore, in (9), assuming that $u = \varphi_s\text{-}V$ and $du = d(\varphi_s\text{-}V)$, the drain current becomes

$$I_{\text{ds}} = \mu_{\text{eff}} W Q_s \frac{dV}{du}\frac{du}{dy}, \quad (12)$$

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and Eq. (3) can be rewritten as:

$$Q_s = C_{OX}(V_{gb} - u - V). \tag{13}$$

Therefore,

$$\frac{dV}{du} = -1 - \frac{1}{C_{OX}} \times \frac{dQ_s}{du}. \tag{14}$$

Combining (12) and (14), the drain current can be expressed as

$$I_{ds0} = -\mu_{eff} W Q_s \frac{du}{dy} - \frac{\mu_{eff} W Q_s}{C_{OX}} \frac{dQ_s}{dy} = I_{ds1} + I_{ds2} \tag{15}$$

where I_{ds1} and I_{ds2} represent diffusion and drift components, respectively. After integration,

$$I_{ds2} = \frac{\mu_{eff} W}{2LC_{OX}}({Q_{ss}}^2 - {Q_{sd}}^2) \tag{16}$$

where L is the channel length. Q_{ss} and Q_{sd} are the charge densities at the source and drain ends, respectively, and they can be acquired from (3) by substituting $\varphi_s = \varphi_{ss}$ and $\varphi_s = \varphi_{sd}$, respectively. Moreover, the diffusion current can be obtained as

$$I_{ds1} = \frac{q\mu_{eff} W}{L} \left\{ \begin{array}{l} \left[\Delta_1 E_1 \left(\exp\left(\frac{u_s}{E_1}\right) - \exp\left(\frac{u_d}{E_1}\right) \right) \right] \\ + \left[\Delta_2 \phi_{th} \left(\exp\left(\frac{u_s}{\phi_{th}}\right) - \exp\left(\frac{u_d}{\phi_{th}}\right) \right) \right] \end{array} \right\} \tag{17}$$

where $u_s = \varphi_{ss}$ and $u_d = \varphi_{sd} - V_{ds}$; $\Delta_1 = N_T \theta_t \exp(\phi_{f0}/E_1)$ and $\Delta_2 = k_B T D_0 \exp(\phi_{f0}/\phi_{th})$. In the meantime, accounting for the influence of non-saturation characteristics,[15] the drain current becomes

$$I_{ds} = (1 + MV_{deff}) I_{ds0} \tag{18}$$

where M is a fitting parameter and $V_{deff} = V_{ds}$-(φ_{sd}-φ_{ss}).

Furthermore, the contact resistance can be significant in MoS_2 FETs.[16,17] As one of the big challenges for realizing a complete model for MoS_2 FETs, the influence of contact resistance should be carefully described. Therefore, we apply a routine method to incorporate contact resistance into the current expression. Firstly, the values of effective contact resistance can be extracted by a DC method[18] as

$$R = \frac{I_{D1} V_{ds2} \left(V_{gst} - \frac{V_{ds2}}{2} \right) - I_{D2} V_{ds1} \left(V_{gst} - \frac{V_{ds1}}{2} \right)}{2 I_{D1} I_{D2} (V_{ds1} - V_{ds2})} \tag{19}$$

where $V_{gst} = V_{gs} - V_T$; I_{D1} and I_{D2} are two different drain currents for a given gate voltage (V_{gs}), corresponding to two drain voltages (V_{ds1} and V_{ds2}) in the linear region. Next, similar to the method of the PSP model,[19] contact resistance is taken into account in the current equation by incorporating into mobility as

$$\mu_{eff_R} = \frac{\mu_{eff}}{1 + G_R}. \tag{20}$$

Here, μ_{eff_R} is the effective mobility considering the contact resistance; and the term $G_R = \alpha \mu_{eff}(W/L) Q_{im} R$ where α is a fitting parameter. Q_{im} is the density of the charge at the potential midpoint and we get it by substituting $\varphi_s = (\varphi_{ss} + \varphi_{sd})/2$ into (3).

In addition, at a high lateral field, the carrier drift velocity begins to saturate. To account for the velocity saturation effect in the developed drain current model, we use a semi-empirical method as Ref. 20

$$I_{ds,cor} = \frac{I_{ds}}{\sqrt{1 + \left(\frac{\mu_{eff_R}}{v_{sat}} E_y \right)^2}} \tag{21}$$

where $I_{ds,cor}$ is the drain current considering the velocity saturation effect; v_{sat} is the saturation velocity, the value of which is 3.4×10^6 cm s^{-1} in our work which is borrowed from Ref. 20; and E_y is the lateral electric field. Using the average value of the electric field $E_y = (\varphi_{sd} - \varphi_{ss})/L$ along the channel, Eq. (21) becomes

$$I_{ds,cor} = \frac{I_{ds}}{\sqrt{1 + \left(\frac{\mu_{eff_R}}{L v_{sat}} (\varphi_{sd} - \varphi_{ss}) \right)^2}}. \tag{22}$$

4. Model verification and discussion

To assess the effectiveness of our scheme, an example of the MoS_2 transistor examined here is shown in Fig. 2, and the manufacturing process can be found in Ref. 8.

To verify the accuracy of our surface potential solution, we complete the following validations in Figs. 3 and 4, with parameters summarized in Table I. The values of D_0 and N_t are borrowed from the reported literature for experimental MoS_2 transistors.[21,22] According to Fig. 3, at the source end, comparisons of surface potential between the explicit calculation of (6) and the iteration results given by (3) are shown. An excellent agreement is achieved, and the maximum absolute error is around 4×10^{-7} V, which improves the accuracy compared with Ref. 7. Moreover, the surface potential with respect to the gate voltage under various channel potentials is shown in Fig. 4, which shows that our surface potential approach is sufficient for the evaluation of the device characteristics.

Furthermore, through the comparisons of three sets of experimental data with different channel lengths, we verify the feasibility of the drain current model expressed by surface potentials. Figure 5 depicts a plot of the modeled drain current versus V_{gs} both in logarithmic and linear scales for the

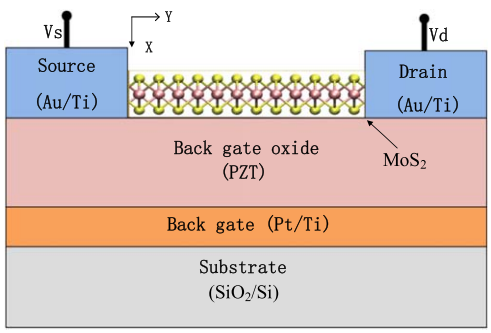

Fig. 2. (Color online) Schematic representation of the PZT gated MoS_2 transistor.

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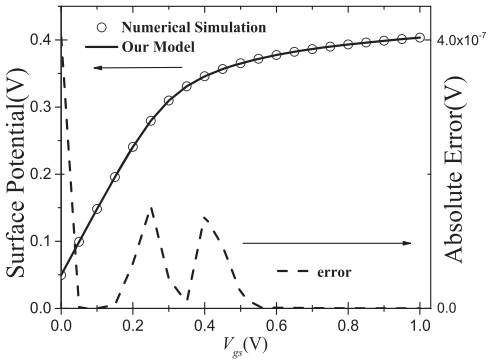

Fig. 3. Comparisons of surface potential solution between numerical results and our model at the source end. Absolute error is also shown.

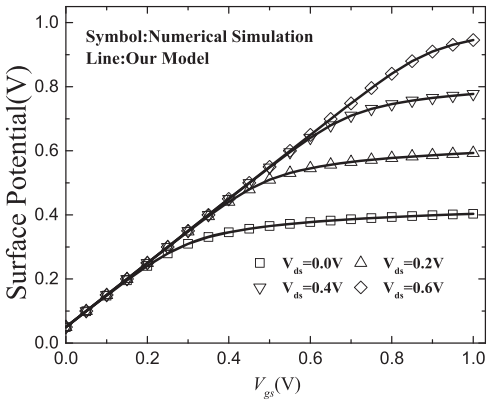

Fig. 4. Comparisons of surface potential solution between numerical results and our model at different drain voltages.

Table I. Parameters used for simulations in Figs. 3–4.

Parameter	Value	Parameter	Value
T [K]	300	C_{OX} [F cm^{-2}]	2.2×10^{-6}
V_{fb}[V]	−0.05	D_0 [cm^{-2}eV^{-1}]	2×10^{14}
ϕ_{f0} [V]	−0.395	N_T[cm^{-2}eV^{-1}]	1.8×10^{13}
T_0 [K]	700		

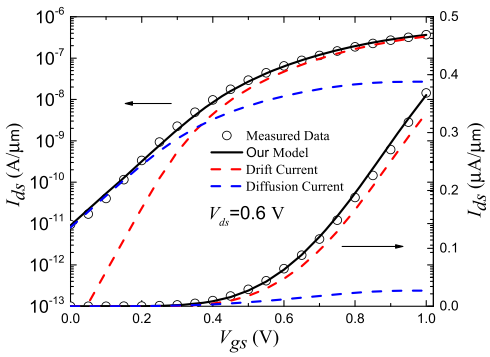

Fig. 5. (Color online) Comparison of modeled (lines) and measured (symbols) transfer characteristics for a device with $L = 4$ μm and $W = 5$ μm at $V_{ds} = 0.6$ V. The process of the symbols is similar to Ref. 8.

device with $L = 4$ μm, and diffusion and drift components are also plotted in this figure. It is not difficult to find that the diffusion and drift currents dominate in the subthreshold and above-threshold regimes, respectively. Figure 6 demonstrates the I_{ds}-V_{ds} characteristics for different applied gate voltages with the parameters presented in Table II. The value of contact resistance is 1.2×10^6 Ω μm, which is obtained from

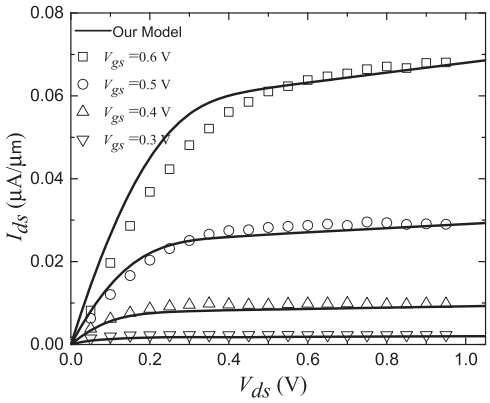

Fig. 6. Comparison of modeled (lines) and measured (symbols) output characteristics for a device with $L = 4$ μm and $W = 5$ μm. The process of the symbols is similar to Ref. 8.

(19), and it is within a reasonable range of resistance values for Ti/Au contact MoS_2 devices.[16,23]

Figures 7 and 8 show that our model accurately predicts the measured transfer and output characteristics of the MoS_2 FETs in different voltage ranges. However, as depicted by the red dotted lines, for such short-channel transistors with $L = 1.5$ μm, the model results of Eq. (18) overestimate the values of drain current, and this can be ascribed to the contact resistance. Therefore, it is necessary to consider contact resistance in the model, and the model combining with formula (20) also shows a better fit with the experimental data. Meanwhile, the comparisons of the transconductance characteristics between the modeled and experimental results are shown in Fig. 9, which further confirms that our model offers continuous and smooth drain current and transconductance characteristics.

In addition, the model fitting of another device from Ref. 24 has been completed in Figs. 10 and 11. The device geometries are: $W/L = 4/7$ μm, the gate oxide thickness is $t_{ox} \sim 50$ nm and the thickness of the MoS_2 layer is $t_{ch} \sim 30$ nm. A good agreement obtained in these figures further verifies the validity of our model.

It should be noted that the model proposed in this article is more suitable for long channel devices. For short-channel devices when ballistic and quasi-ballistic transports should be taken into account, we can use the delta function[9] or virtual source model.[25] Theoretically, this model is also valid for other transition metal dichalcogenide based thin body transistors like WSe_2 and $MoTe_2$ with modifications of the values of TMD material parameters such as bandgap, effective mass, and dielectric constant.

5. Conclusion

In this paper, a complete explicit model considering the band-tail effect for surface potential has been presented which has the advantages of both accuracy and computational efficiency. Furthermore, combining contact resistance and velocity saturation effect, a physics-based drain current model is developed. Its validity has been verified by experimental results with different bias and channel length conditions. This physics-based model is expected to be useful for electronic devices made by this new material.

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Table II. Parameters used for simulations.

Symbol (unit)	Value in Figs. 5 and 6	Value in Figs. 7–9	Value in Figs. 10–11
C_{OX}(F cm^{-2})	2.2×10^{-6}	2.2×10^{-6}	1.59×10^{-7}
ϕ_{f0} (V)	−0.39	−0.395	−0.31
D_0 (cm^{-2}eV^{-1})	2×10^{14}	2×10^{14}	1×10^{14}
N_T (cm^{-2}eV^{-1})	2×10^{13}	1.8×10^{13}	3×10^{12}
T_0 (K)	650	700	2000
V_{fb} (V)	−0.05	−0.05	−1.7
R (Ω μm)	1.2×10^{6}	2×10^{6}	6.8×10^{6}
M (−)	0.2	0.13	0

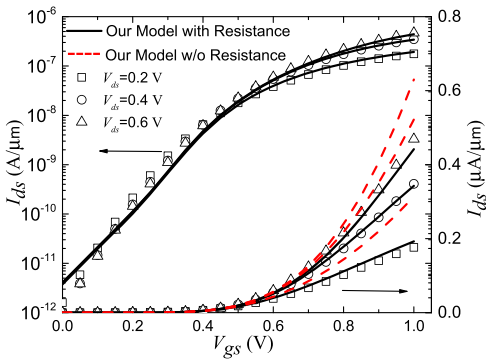

Fig. 7. (Color online) Comparison of modeled (lines) and measured (symbols)[8] transfer characteristics for a device with $L = 1.5$ μm and $W = 20$ μm.

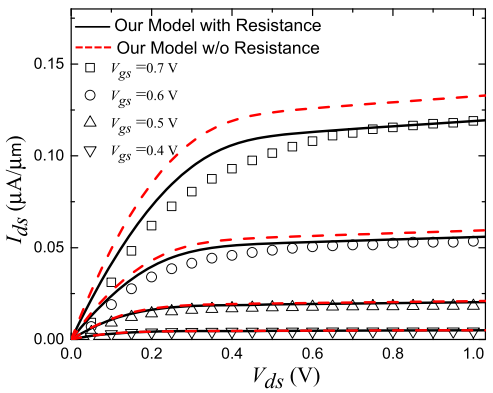

Fig. 8. (Color online) Comparison of modeled (lines) and measured (symbols)[8] output characteristics for a device with $L = 1.5$ μm and $W = 20$ μm.

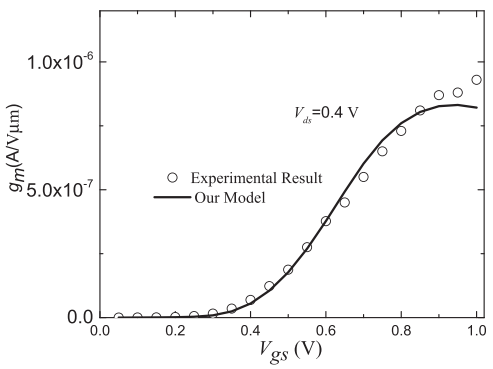

Fig. 9. Comparison of transconductance g_m between modeled results and experimental data[8] with $W/L = 20$ μm $L^{-1} = 1.5$ μm.

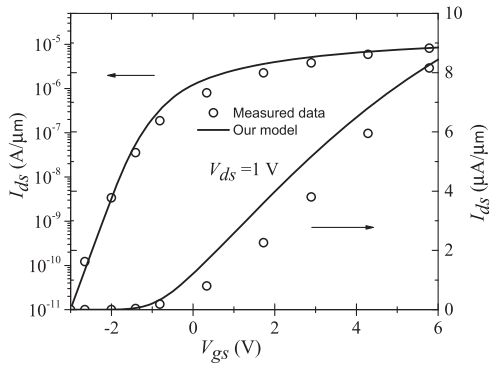

Fig. 10. Comparison of modeled (lines) and measured (symbols)[24] transfer characteristics for a device with $L = 7$ μm, $W = 4$ μm, $t_{ox} = 50$ nm and $t_{ch} = 30$ nm.

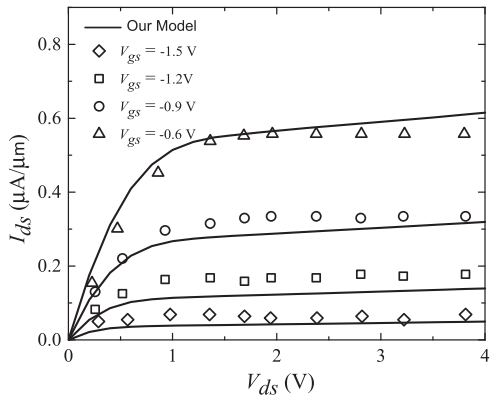

Fig. 11. Comparison of modeled (lines) and measured (symbols)[24] output characteristics for a device with $L = 7$ μm, $W = 4$ μm, $t_{ox} = 50$ nm and $t_{ch} = 30$ nm.

Acknowledgments

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